Ethics of AI in Healthcare

Sophie Rogers
S.Rogers@kent.ac.uk

Overview

- Types of medical AI systems.
- Current laws and regulations for medical AI systems.
- Ethical barriers to adoption of AI in the NHS.
- Case studies and current application in the NHS.

Types of Medical AI systems

Clinical, Pharmaceutical, Administrative

- Medical Imaging and Diagnostics
- Robotics in Surgery

- Drug Discovery and Development

- NLP Speech Recognition
- CDSS (Clinical Decision Support Systems)

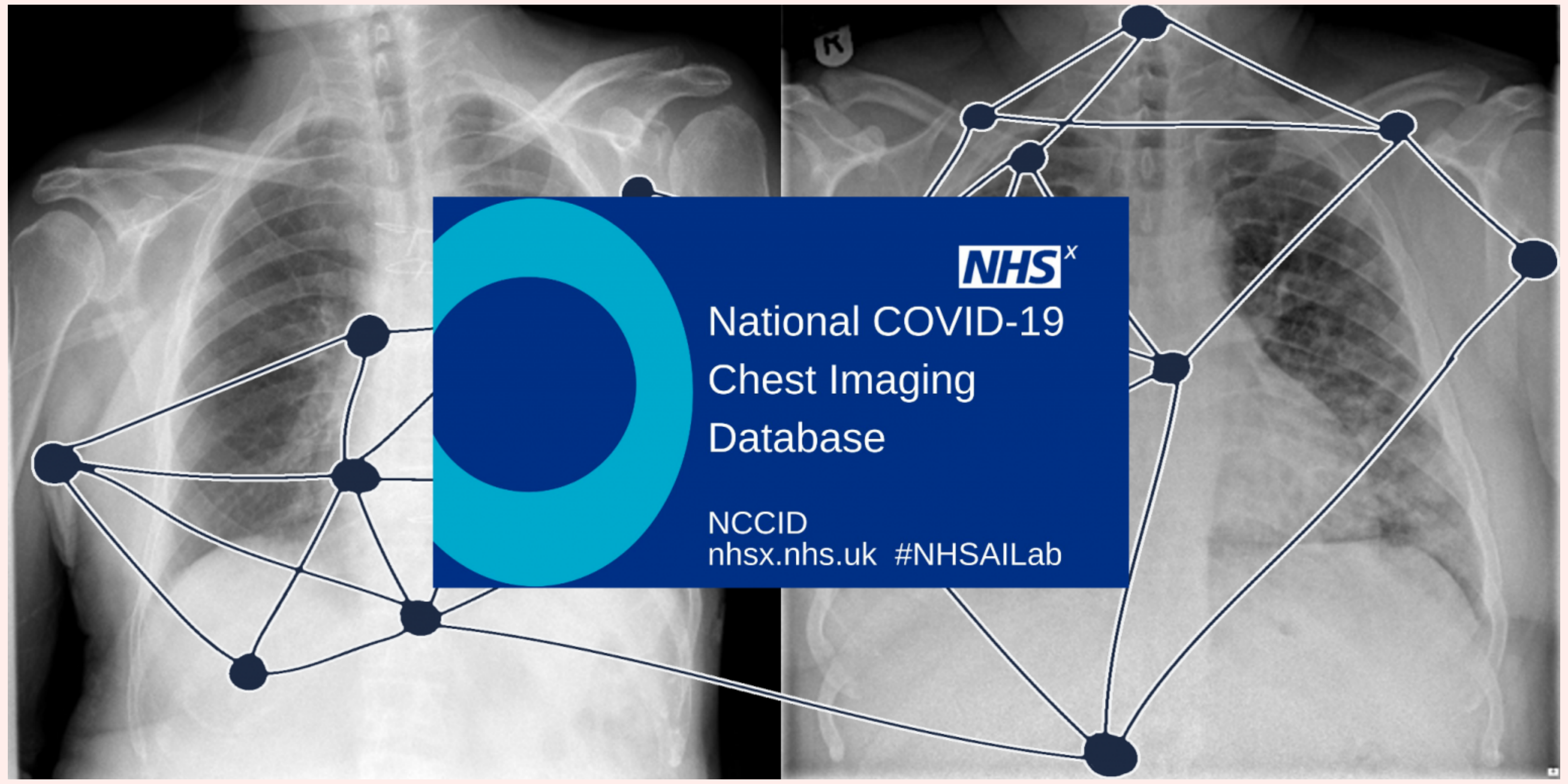
Clinical AI systems

Medical Imaging and Diagnostics

- National COVID-19 Chest Imaging Database (NCCID)

Robotics in Surgery

- Guy's and St Thomas' Surgical Robotics Programme - 13 years



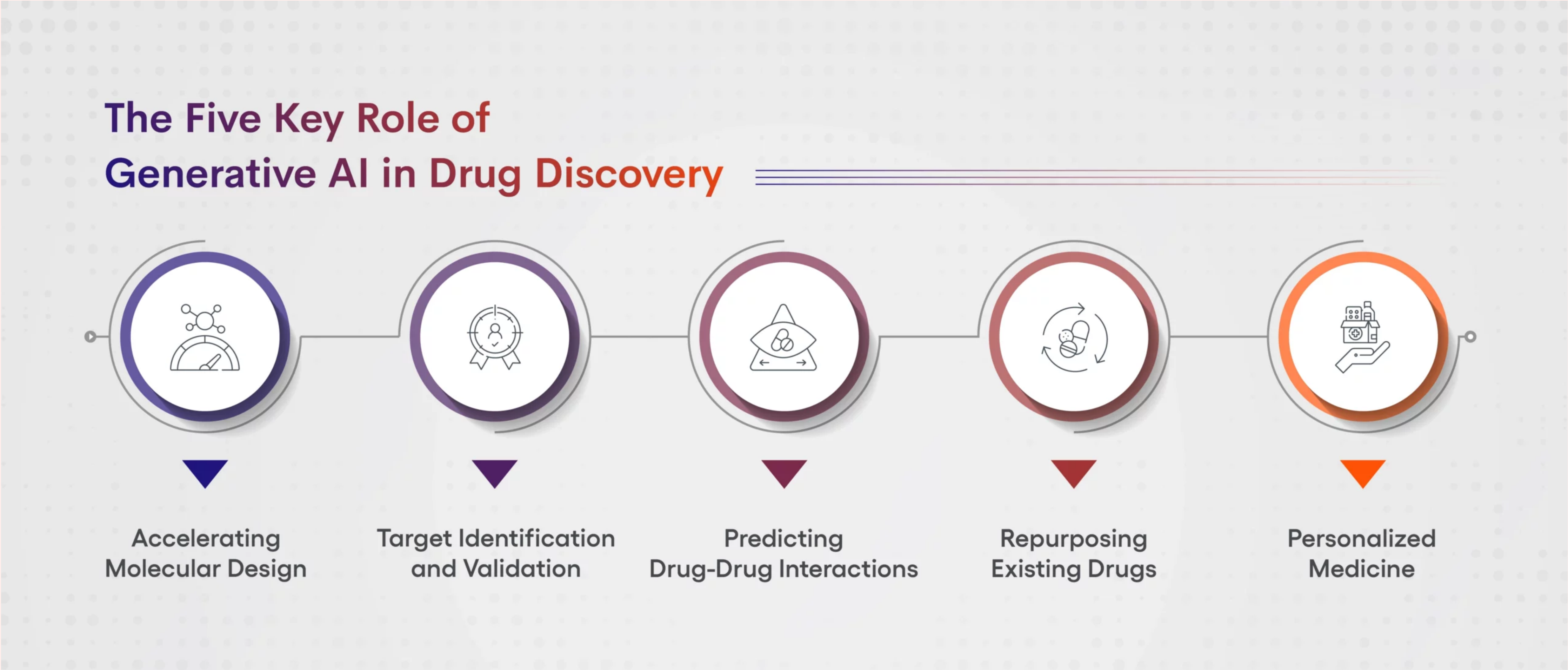
Pharmaceutical **AI systems**

Drug Discovery and Development

- Improves speed and reduces costs

Automating Clinical Trials

- Reduce cycle times and costs, while also improving the outcomes of clinical development.



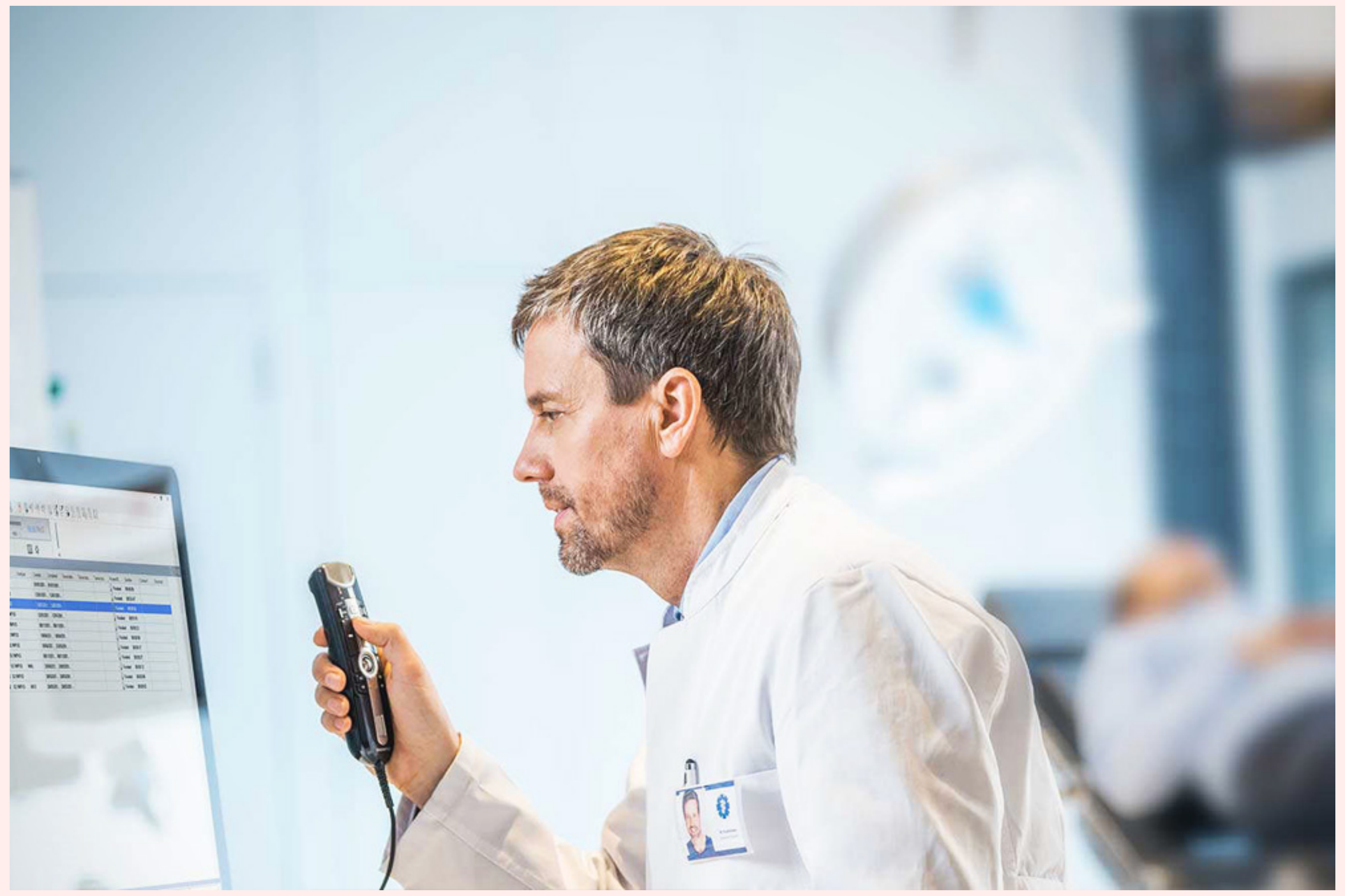
Administrative AI systems

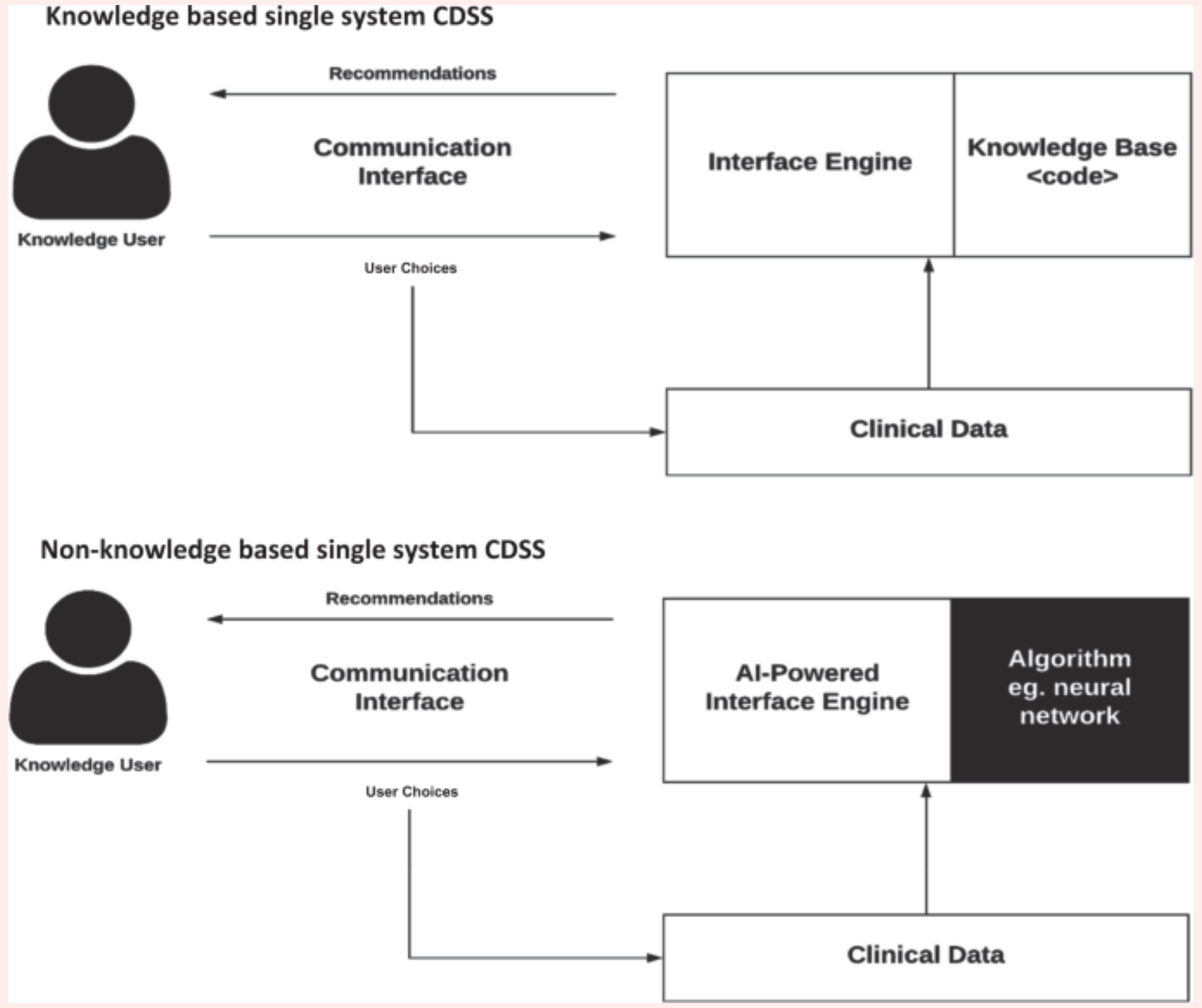
CDSS (Clinical Decision Support Systems)

- Knowledge based or non-knowledge based

Clinical Notes

- NLP Speech Recognition
- NLP Text Summarisation





Current laws and regulations for medical AI systems.

UK Framework for AI Regulation

The UK Government has adopted a cross-sector and outcome-based framework for regulating AI, underpinned by five core principles.

UK Framework for AI Regulation

- Accountability and Governance
- Fairness
- Safety, Security and Robustness
- Appropriate Transparency and Explainability
- Contestability and Redress

Current laws and regulations for medical AI systems.

NHS AI and Digital Regulations Service

The AI and Digital Regulations Service is a multi-agency collaboration between 4 organisations.

The National Institute for Health and Care Excellence

Evaluate the clinical and cost effectiveness of health technologies and produce evidence-based guidance and advice for health, public health and social care.

The Medicines and Healthcare products Regulatory Agency

Regulate medicines and medical devices, including software as a medical device.

The Health Research Authority

Protect the rights of patients, including by regulating the use of data collected within health and social care for research and product development.

The Care Quality Commission

Monitor, inspect and regulate services to make sure they meet fundamental standards of quality and safety and publish findings.

Ethical barriers to adoption.

- AI has the potential to revolutionise healthcare
- More than half of all of the CE marked AI devices developed between 2015 and 2020 were intended for use in diagnostic imaging.

So why isn't it in use?

Ethical barriers to adoption.

Algorithm Bias

Transparency and Explainability

So why isn't it in use?

Lack of Trust

Privacy Concerns

Need for large volumes of data.

Algorithmic Bias

AI bias:

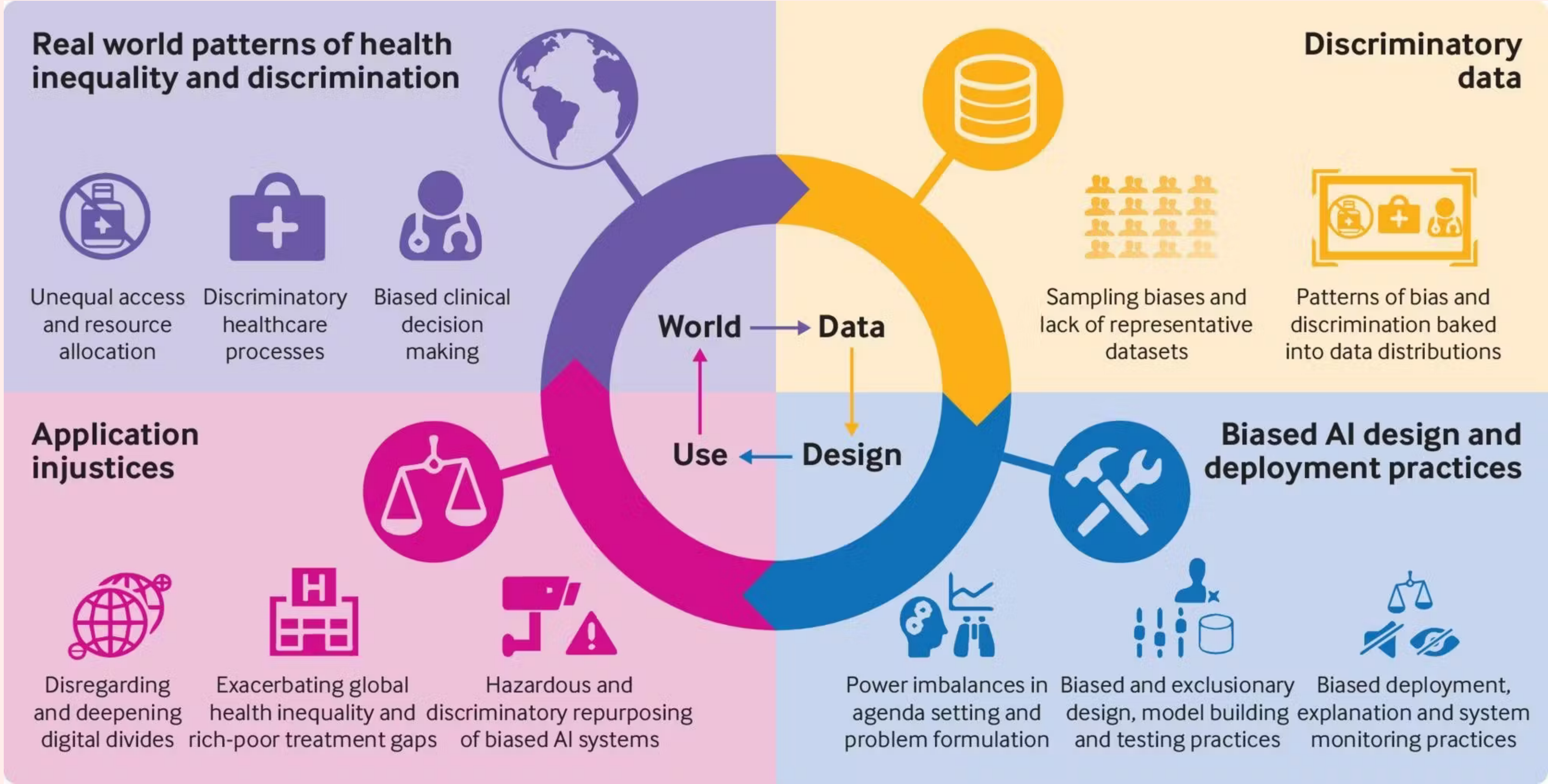
Also referred to as algorithmic bias or machine learning bias, occurs when algorithms make decisions which either favour, or disadvantage, certain groups of people due to human biases impacting the training data or algorithm.

Gender bias revealed in AI tools screening for liver disease

11 July 2022

Artificial intelligence models built to predict liver disease from blood tests are twice as likely to miss disease in women as in men, a new study by UCL researchers has found.

If the training data is not representative of the general population, AI models can be biased and amplify and reinforce existing inequalities in medicine.


Real world patterns of health inequality and discrimination
Unequal access and resource allocation
Discriminatory healthcare processes
Biased clinical decision making
Discriminatory data
Sampling biases and lack of representative datasets
Patterns of bias and discrimination baked into data distributions
World
Data
Design
Use
Application injustices
Disregarding and deepening digital divides
Exacerbating global health inequality and rich-poor treatment gaps
Hazardous and discriminatory repurposing of biased AI systems
Biased AI design and deployment practices
H
Power imbalances in agenda setting and problem formulation
Biased and exclusionary design, model building and testing practices
Biased deployment, explanation and system monitoring practices

Transparency and Explainable AI (XAI)

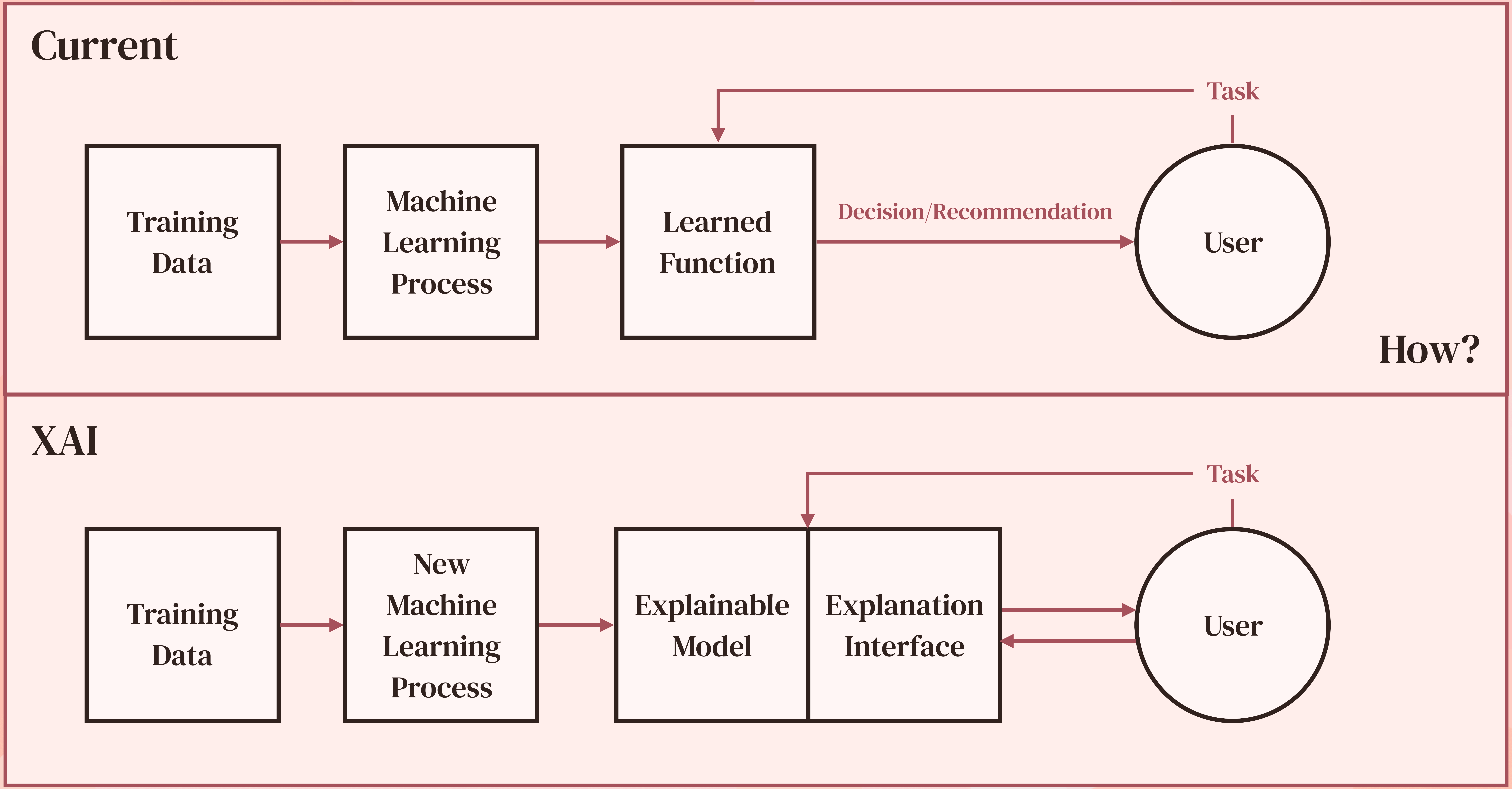
Explainability:

The concept that a machine learning model and its output can be explained in a way that “makes sense” to a human being at an acceptable level.

Transparency:

Understanding how artificial intelligence systems make decisions, why they produce specific results, and what data they’re using.

- AI models are often seen as black boxes.
- In healthcare, the aim is Explainable Artificial Intelligence (XAI).

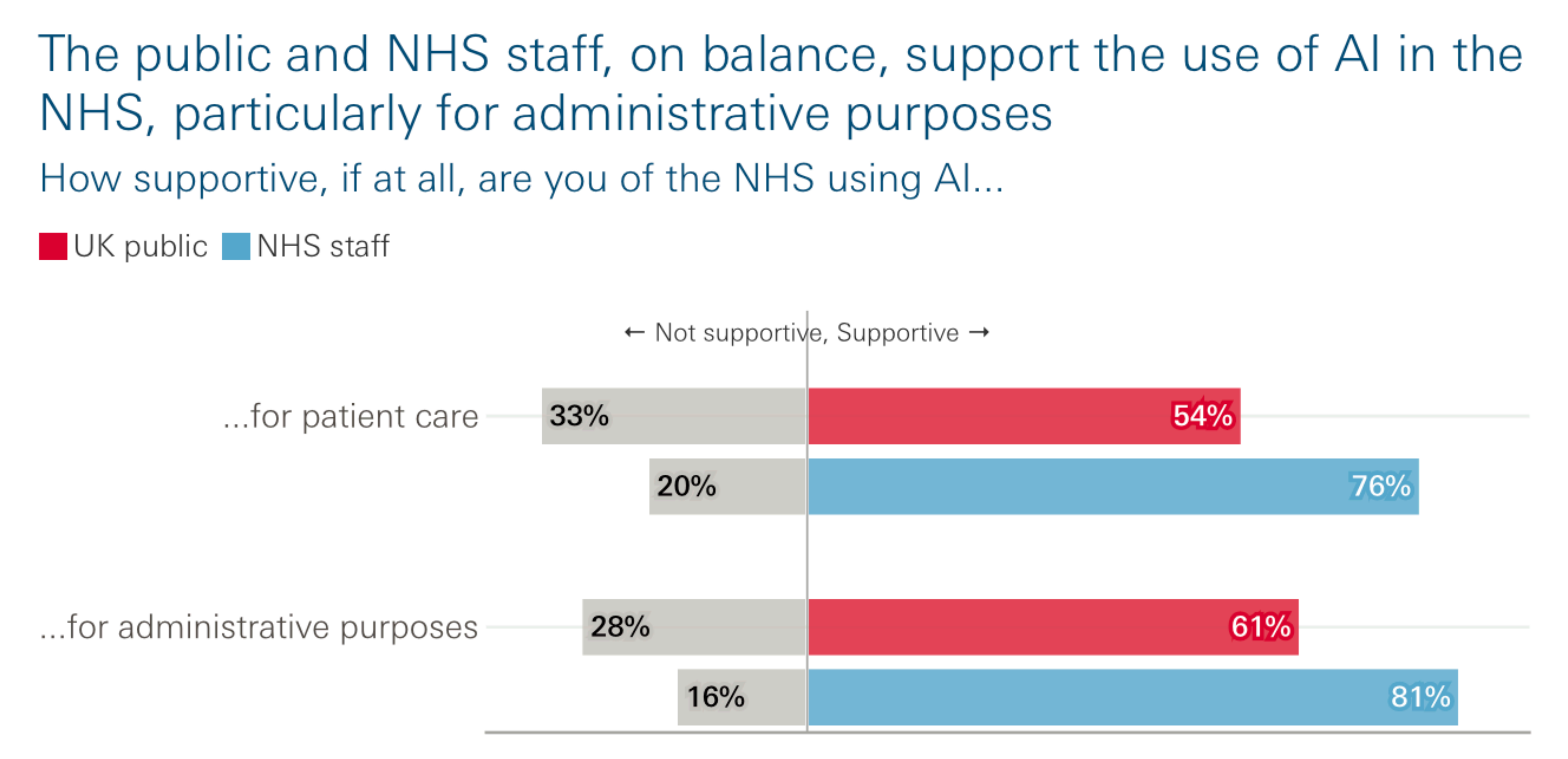

Current
Training Data
Machine Learning Process
Learned Function
Task
Decision/Recommendation
User
How?
XAI
Training Data
New Machine Learning Process
Explainable Model
Explanation Interface
Task
User

Trust in Artificial Intelligence

2022 NHS AI Lab and Health Education England Collaboration

Developing healthcare workers' confidence in artificial intelligence (AI) Report.

'the challenge is to provide the right resources to the right people. This means providing a solid foundation for developing AI-related knowledge as well as personalised advanced educational elements to fit the needs of individuals in different roles and responsibilities'.



I think AI will...

Improve Care Quality

Not change Care Quality

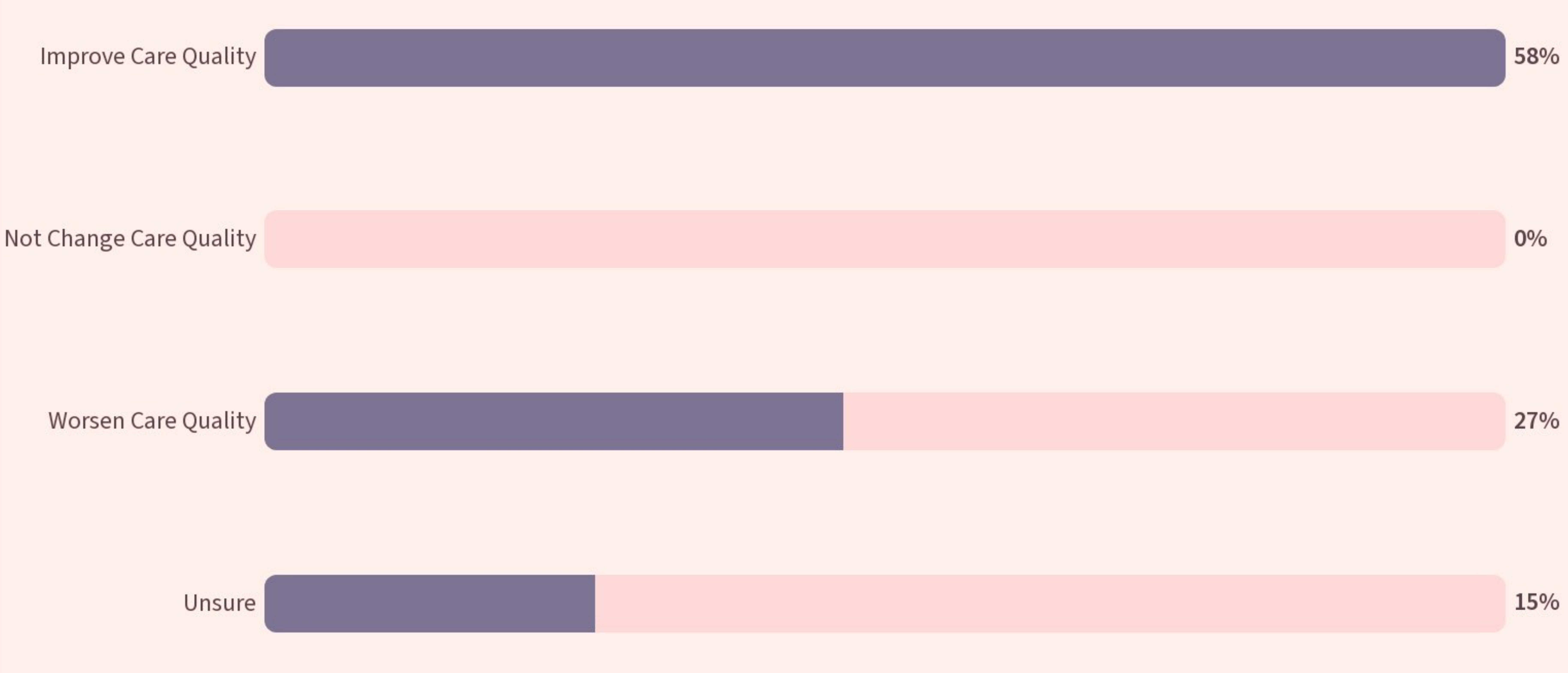
Worsen Care Quality

Unsure

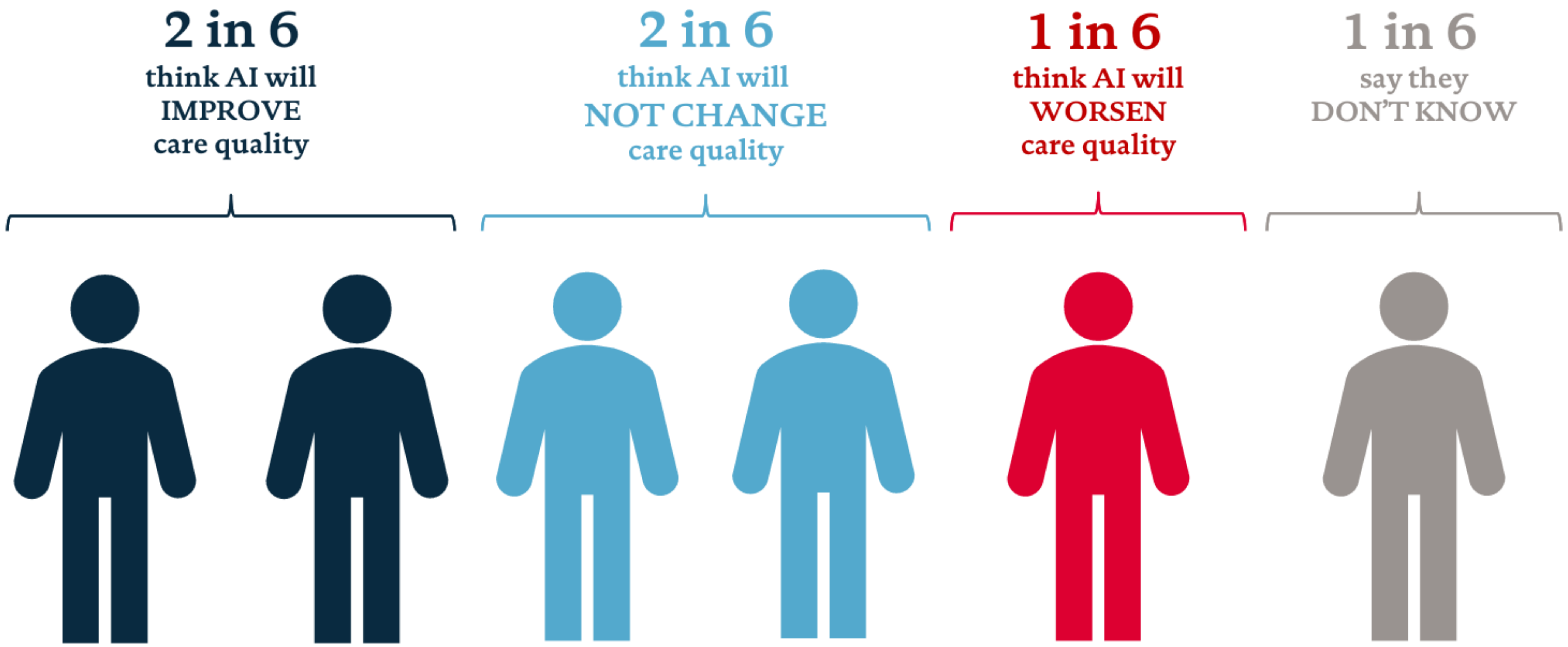
QR Code for live poll voting no longer live.

Live Updating Results Slide

I think AI will…



The need for public engagement: 1 in 6 think AI will make care quality worse



The Health Foundation

Exact percentages were: 'I think that using AI will make the quality of health care better' (33.2%), 'I think that using AI will not change the quality of health care' (32.9%), 'I think that using AI will make the quality of health care worse' (17.7%), 'Don't know' (16.2%).

Privacy

- AI systems require large amounts of training data – where is this coming from?
- Non-Medical organisations beginning to give consumers a choice in how their data is used.
- KPMG 2024 – 63% of consumers have concerns about misuse of their data via generative AI.
- Meta AI 2024
- How are the NHS consenting their patients, and how are they using their data?

Learn how we use your information as we expand AI at Meta

We're getting ready to expand our AI at Meta experiences. AI at Meta is our collection of generative AI features and experiences, such as Meta AI and AI creative tools, along with the models that power them.

Why your data is important

The NHS uses information about patients (patient data) to research, plan and improve:

- the services we offer
- the treatment and care patients receive

We get this data from your GP surgery, hospitals and other healthcare providers. The organisation that collects your data is called NHS England.

To help improve services, NHS England shares this data with researchers from organisations such as universities or hospitals. This type of data-sharing has been happening for many years.

All data that is collected and shared is protected by strict rules around privacy, confidentiality and security.

Automated Decision Making

- Early stages of implementation, largely experimental and used in a research context.
- Augmented decision-making, rather than entirely AI-driven.
- Rostering staff within the NHS (Limited staff data only)
- Improving administrative efficiency (no personal data).
- UK GDPR Article 22
- Human-Machine Interaction and Deciding Whether or not to Treat.

Case studies and current application in the NHS.

NEWS

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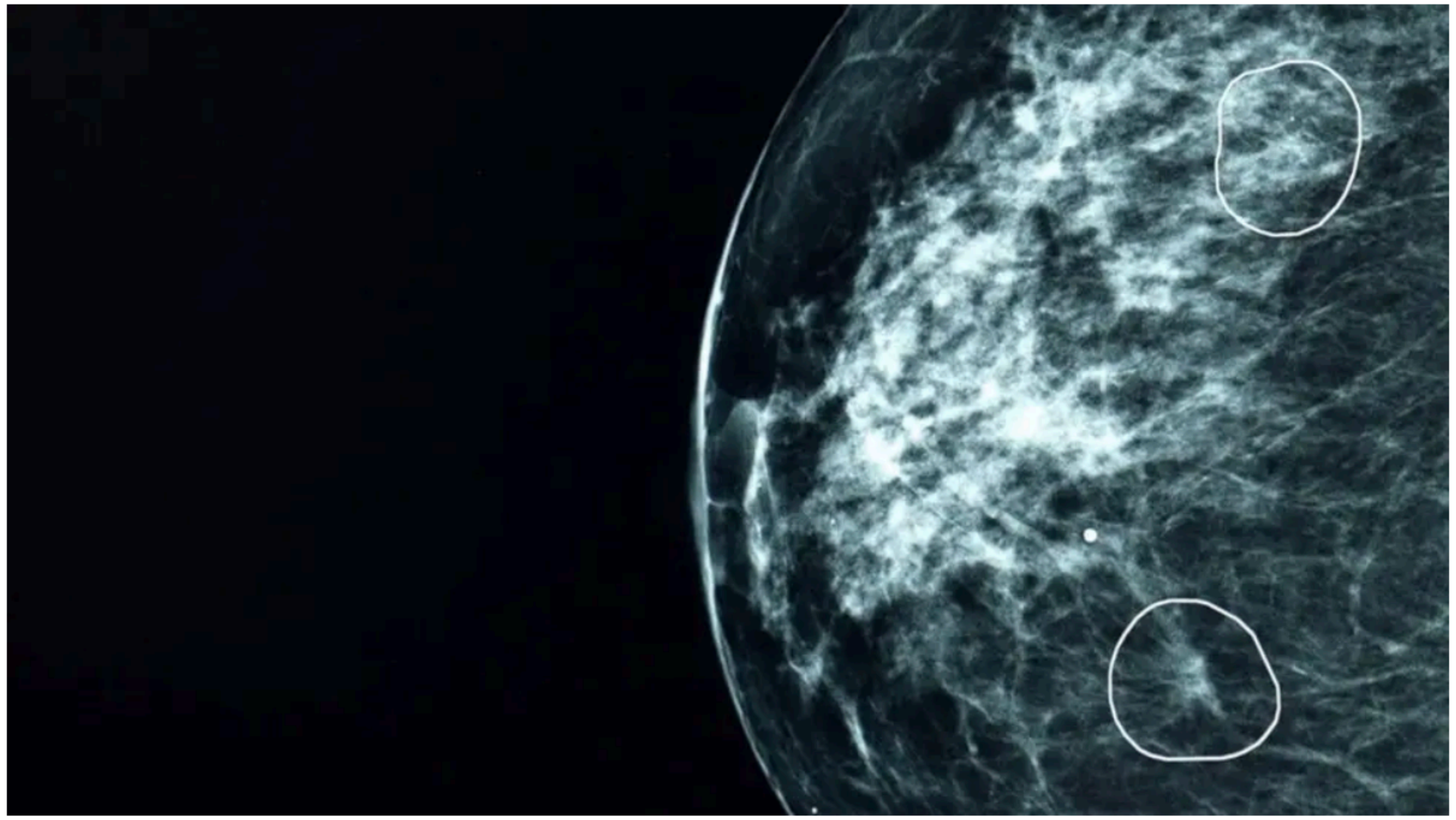
Technology

NHS AI test spots tiny cancers missed by doctors

21 March



AI tool Mia circles two areas of concern on a mammogram scan

Stroke patient says AI diagnosis was a 'lifeline'

Mr Theoff (r) said he feels "lucky" to have recovered without any serious issues

Joshua Askew
BBC News, South East

4 October 2024

News

NHS AI expansion to help tackle missed appointments and improve waiting times

14 March 2024

Innovation

The NHS is set to roll out artificial intelligence (AI) to reduce the number of missed appointments and free up staff time to help bring down the waiting list for elective care.

The expansion to ten more NHS Trusts follows a successful pilot in Mid and South Essex NHS Foundation Trust, which has seen the number of did not attends (DNAs) slashed by almost a third in six months.

Using an AI Chatbot to ease staff shortages and improve patient care

Faced with growing demand and limited resources, the mental health team of an NHS trust identified a critical issue within their talking-therapy service.

Published at 13 March 2024 by AI and Digital Regulations Service for health and social care

Adopters Chatbots